

torch.tensor_split#

https://pytorch.org/docs/stable/generated/torch.tensor_split.html#torch.tensor_split

Description:

Splits a tensor into multiple sub-tensors, all of which are views of input, along dimension `dim` according to the indices or number of sections specified by `indices_or_sections`. This function is based on NumPy's `numpy.array_split()`.

input (Tensor) – the tensor to split

indices_or_sections (Tensor, int or list or tuple of ints) – If `indices_or_sections` is an integer `n` or a zero dimensional long tensor with value `n`, input is split into `n` sections along dimension `dim`. If input is divisible by `n` along dimension `dim`, each section will be of equal size, `input.size(dim) / n`. If input is not divisible by `n`, the sizes of the first `int(input.size(dim) % n)` sections will have size `int(input.size(dim) / n) + 1`, and the rest will have size `int(input.size(dim) / n)`. If `indices_or_sections` is a list or tuple of ints, or a one-dimensional long tensor, then input is split along dimension `dim` at each of the indices in the list, tuple or tensor. For instance, `indices_or_sections=[2, 3]` and `dim=0` would result in the tensors `input[:2]`, `input[2:3]`, and `input[3:]`.

Function Signature

```
torch.tensor_split(input, indices_or_sections, dim=0) → List  
of Tensors#
```

Parameters

Code Examples

```
>>> x = torch.arange(8)
>>> torch.tensor_split(x, 3)
(tensor([0, 1, 2]), tensor([3, 4, 5]), tensor([6, 7]))
```

```
>>> x = torch.arange(7)
>>> torch.tensor_split(x, 3)
(tensor([0, 1, 2]), tensor([3, 4]), tensor([5, 6]))
>>> torch.tensor_split(x, (1, 6))
(tensor([0]), tensor([1, 2, 3, 4, 5]), tensor([6]))
```

```
>>> x = torch.arange(14).reshape(2, 7)
>>> x
tensor([[ 0,  1,  2,  3,  4,  5,  6],
        [ 7,  8,  9, 10, 11, 12, 13]])
>>> torch.tensor_split(x, 3, dim=1)
(tensor([[0, 1, 2],
        [7, 8, 9]]),
 tensor([[3, 4],
        [10, 11]]),
 tensor([[5, 6],
        [12, 13]]))
>>> torch.tensor_split(x, (1, 6), dim=1)
(tensor([[0],
        [7]]),
 tensor([[1, 2, 3, 4, 5],
        [8, 9, 10, 11, 12]]),
 tensor([[6],
        [13]]))
```

Detailed Documentation

Documentation

Splits a tensor into multiple sub-tensors, all of which are views of input, along dimension `dim` according to the indices or number of sections specified by `indices_or_sections`. This function is based on NumPy's `numpy.array_split()`.

`input` (Tensor) – the tensor to split

`indices_or_sections` (Tensor, int or list or tuple of ints) –

If `indices_or_sections` is an integer `n` or a zero dimensional long tensor with value `n`, `input` is split into `n` sections along dimension `dim`. If `input` is divisible by `n` along dimension `dim`, each section will be of equal size, `input.size(dim) / n`. If `input` is not divisible by `n`, the sizes of the first `int(input.size(dim) % n)` sections will have size `int(input.size(dim) / n) + 1`, and the rest will have size `int(input.size(dim) / n)`.

If `indices_or_sections` is a list or tuple of ints, or a one-dimensional long tensor, then `input` is split along dimension `dim` at each of the indices in the list, tuple or tensor. For instance, `indices_or_sections=[2, 3]` and `dim=0` would result in the tensors `input[:2]`, `input[2:3]`, and `input[3:]`.

If `indices_or_sections` is a tensor, it must be a zero-dimensional or one-dimensional long tensor on the CPU.

`dim` (int, optional) – dimension along which to split the tensor. Default: 0

